

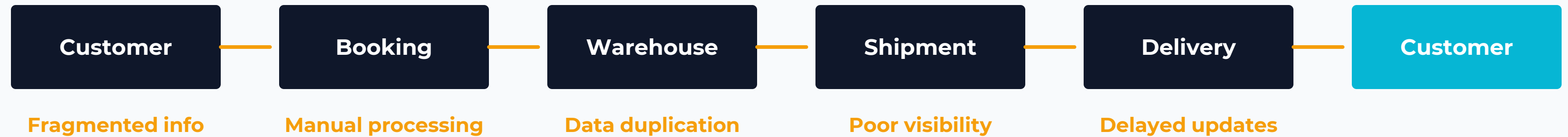
ACADEMIC DBMS PROJECT • 2025

ENTERPRISE PARCEL MANAGEMENT SYSTEM

Database Management System — Project Implementation

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Problem Statement



Disconnected records make every handoff harder to verify — slowing operations, weakening accountability, and limiting reliable reporting.

MANUAL RECORDS • INCONSISTENT DATA • AGENT & WAREHOUSE COMPLEXITY • LIMITED REPORTING

Project Objectives

01

Centralized Database

02

Parcel Tracking

03

Data Integrity

04

Efficient
Management

05

Secure Access

06

Fast Data Retrieval

07

Reporting

08

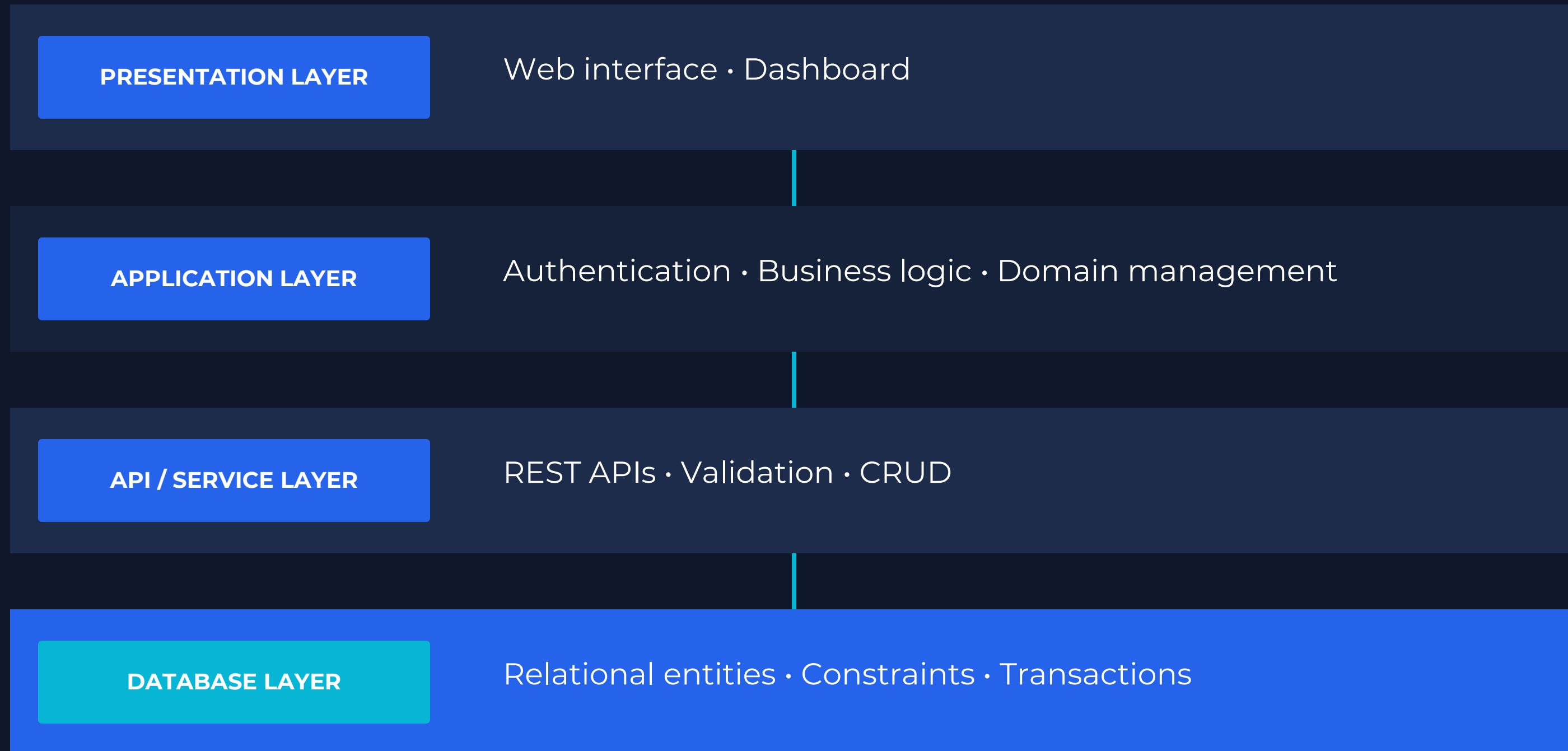
Scalability

Proposed System



One source of truth coordinates every operational handoff — from booking and inventory to delivery confirmation and payment.

System Architecture



Modules & Functionalities

Customer

Register / login
Manage profile
Track parcels

Admin

Manage users
Roles & access
Operational reports

Parcel

Create record
Edit details
Status history

Shipment

Plan shipment
Assign warehouse
Update movement

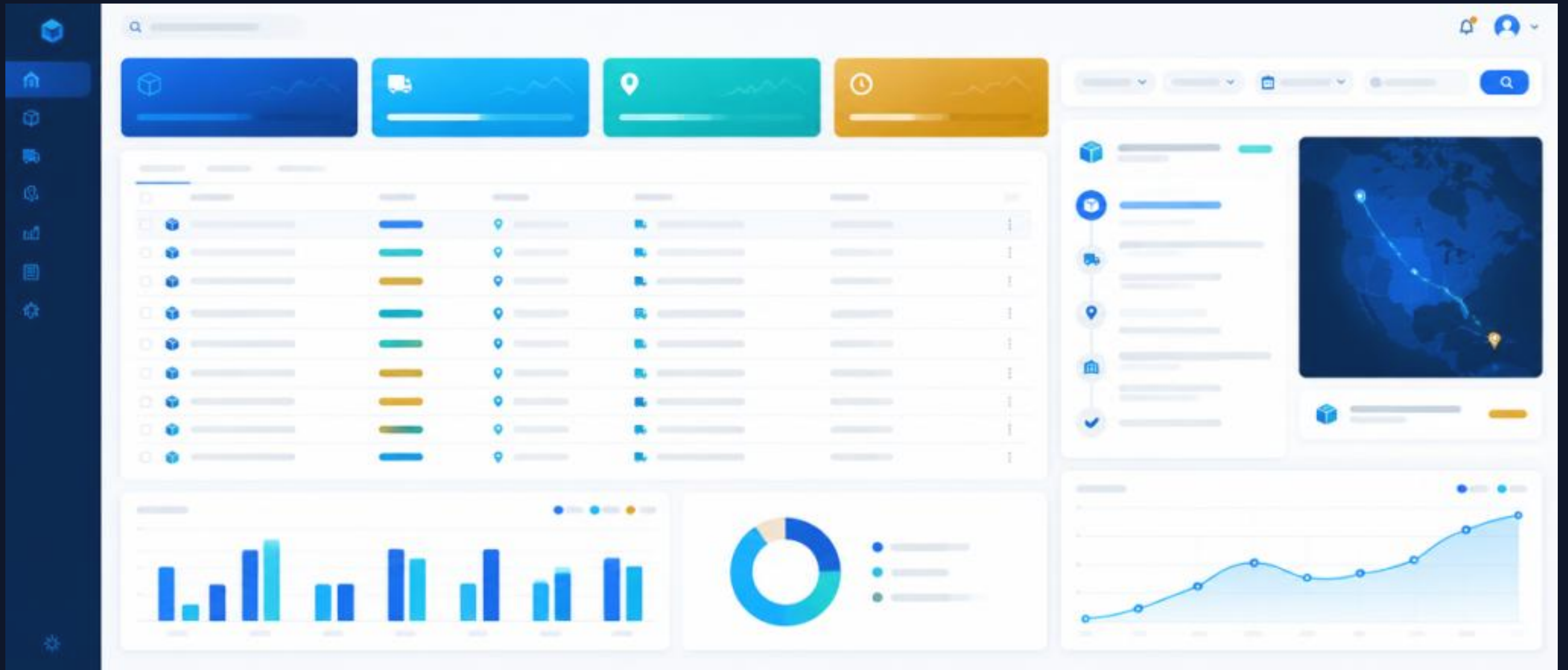
Delivery

Assign agent
Proof of delivery
Completion status

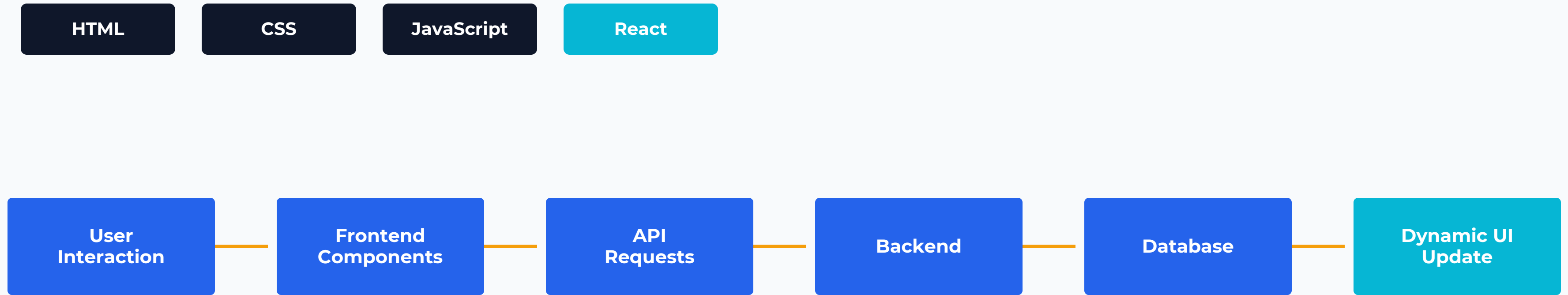
Payment

Record payment
Verify transaction
Payment history

Frontend Design



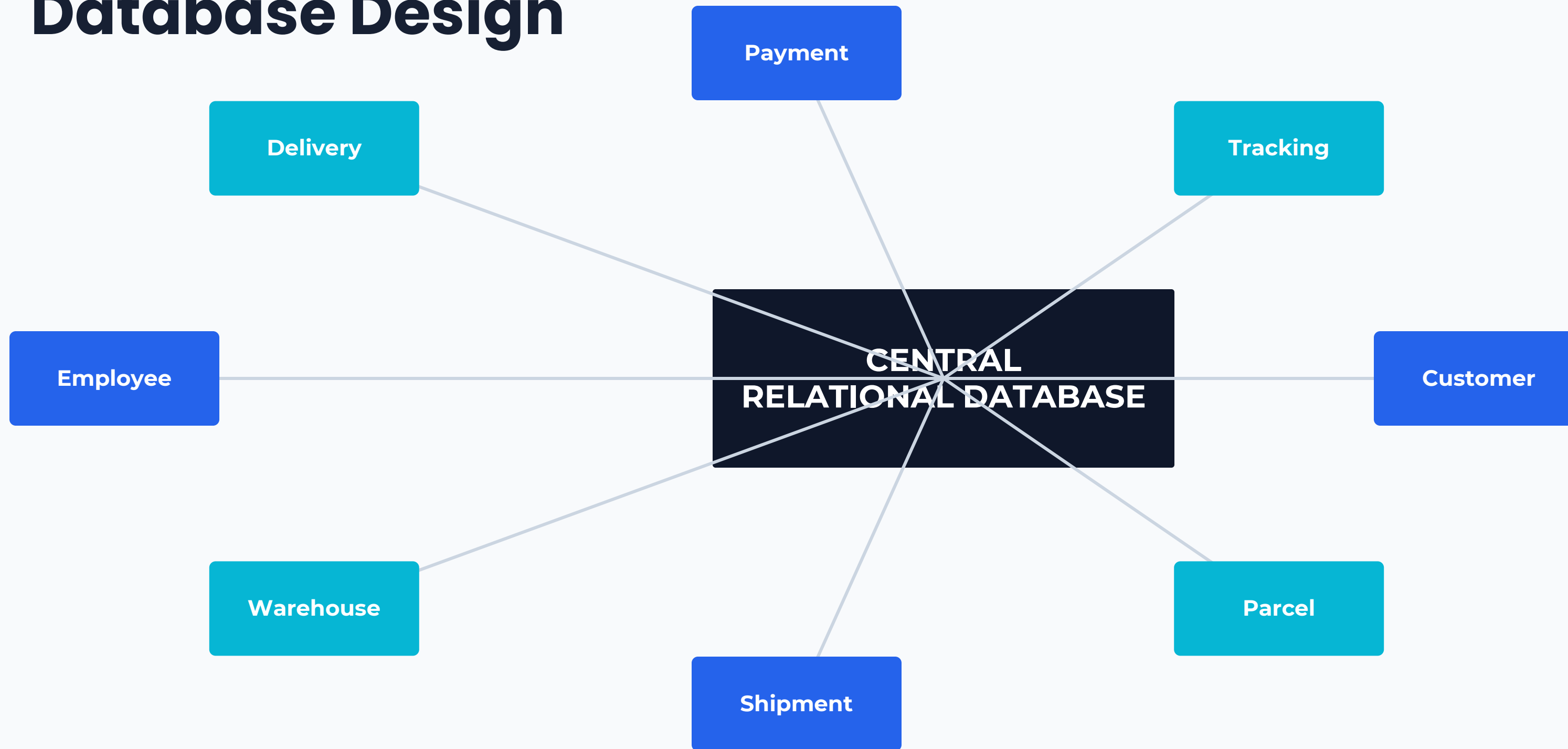
Frontend Implementation



Responsive navigation • CRUD forms • Search & filters • Authentication • API integration • Client-side validation

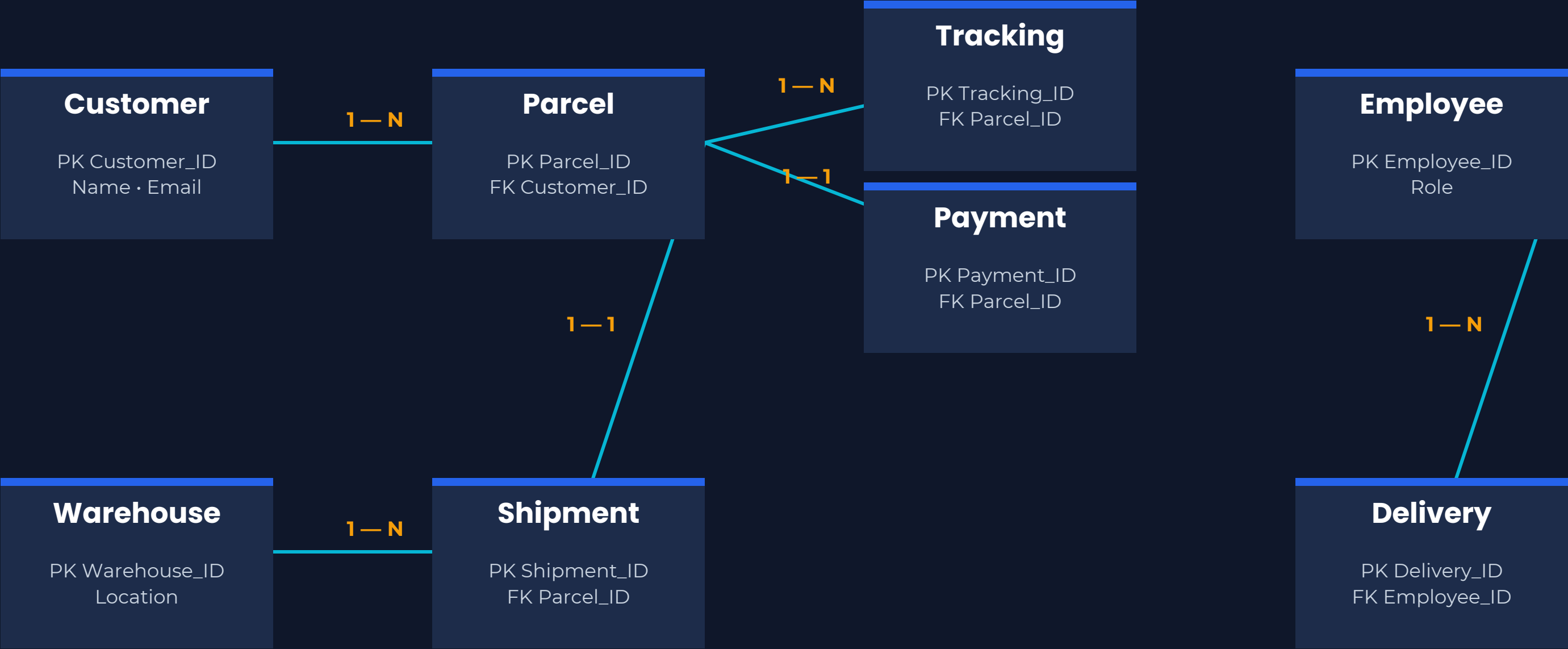
A clean component model converts validated user actions into reliable database transactions — then refreshes the interface with current state.

Database Design



PRIMARY & FOREIGN KEYS • NORMALIZATION • CONSTRAINTS • REFERENTIAL INTEGRITY • SQL CRUD

Entity Relationship Diagram



Database Tables & Schema

CUSTOMER	PARCEL	SHIPMENT
Customer_ID INT PK Name VARCHAR(100) Email VARCHAR(120) Phone VARCHAR(15)	Parcel_ID INT PK Customer_ID INT FK Source VARCHAR(80) Destination VARCHAR(80) Status VARCHAR(30)	Shipment_ID INT PK Parcel_ID INT FK Warehouse_ID INT FK Ship_Date DATE Status VARCHAR(30)

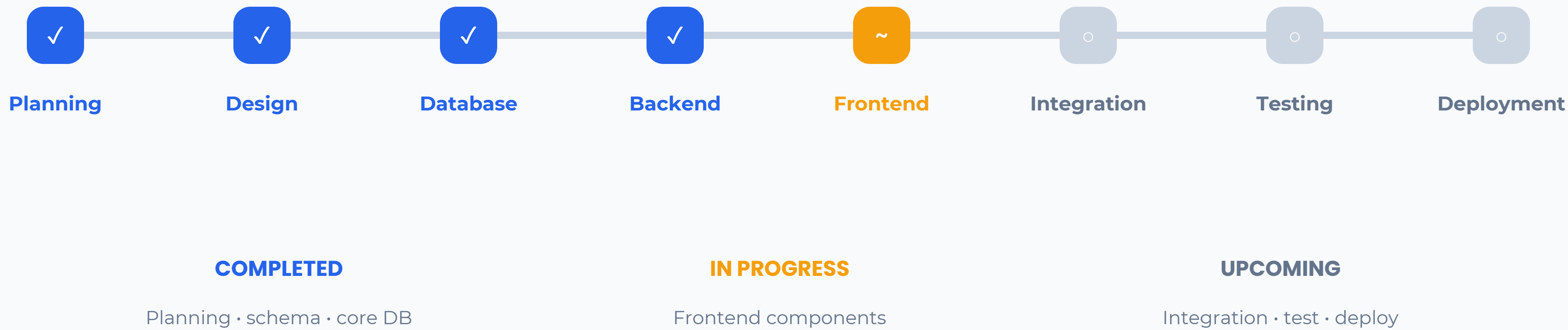
Sample Database Records

CUSTOMER				
ID	Name	Email	Phone	
C101	Aanya Rao	aanya@sample.in	98765 21430	
C102	Rohan Mehta	rohan@sample.in	99870 44218	

PARCEL				
ID	Customer	Source	Destination	Status
P501	C101	Pune	Bengaluru	IN TRANSIT
P502	C102	Mumbai	Hyderabad	BOOKED

SHIPMENT			
ID	Parcel	Warehouse	Status
S301	P501	W04	DISPATCHED
S302	P502	W02	PROCESSING

Current Project Progress



Challenges & Future Work

CURRENT CHALLENGES

- 01 Database relationships
- 02 Data integrity
- 03 CRUD integration
- 04 Validation
- 05 Scalability
- 06 Synchronization

FUTURE WORK

- 01 Real-time GPS tracking
- 02 Route optimization
- 03 SMS / email / push
- 04 AI-powered analytics
- 05 Advanced dashboard
- 06 Cloud deployment

CONCLUSION

ONE CONNECTED SYSTEM.

Customers, parcels, shipments, warehouses, delivery, payments, and tracking — unified through structured relational data and dependable CRUD workflows.

STRUCTURED DATA + RELATIONSHIPS + CRUD + FRONTEND + BACKEND

“From Parcel Booking to Final Delivery — One Connected System.”